

CHENHAO ZHOU

Ph.D. Candidate | Instructor of Record

Department of Supply Chain Management | Rutgers Business School

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EDUCATION

Rutgers Business School, Rutgers University

Expected May 2027

Ph.D. in Management | Concentration: Supply Chain Management | GPA: 4.00/4.00

Dissertation defense scheduled May 20, 2027

Advisors: [David Dreyfus](#) • [Lei Lei](#) (Early-Stage)

Dissertation Committee: [Weiwei Chen](#), [Xin Ding](#), [Keith Skowronski](#)

Tandon School of Engineering, New York University

2021

M.S. in Engineering / Industrial Management

Questrom School of Business, Boston University

2019

B.S. in Business Administration

RESEARCH INTERESTS

Healthcare operations and workforce management • Buyer-supplier relationships and innovation • Public policy and environmental sustainability in operations • AI placement in service processes and causal inference methods

TECHNICAL SKILLS

Programming

Python, R, SQL, Stata, MATLAB, L^AT_EX

Optimization & Simulation Software

AnyLogic, AMPL, CPLEX, Gurobi, OptQuest

Econometrics

Panel data methods, instrumental variables, control function, local projections, structural estimation

Machine Learning

Causal ML (double/debiased ML, causal forests), deep reinforcement learning

Visualization

Power BI, Tableau, HTML, React, JavaScript

DISSERTATION

Title: How Health Care Delivery Organizations Absorb External Change: Workforce, Policy, and Technology.

Defense scheduled May 20, 2027.

The dissertation asks how delivery organizations absorb external change of three kinds and finds the same two conditions governing each: whether the organization is operating under high demand, and how far the affected groups differ in their capacity to adapt. Read together the essays rank three responses to that bind, from weakest to strongest: a single institution's own operating decisions, system-level policy support, and placing technology where it actually binds. The published oncology workflow study is the base point from which the essays radiate.

Four papers: an opening chapter published in *Annals of Operations Research* [1], and three essays under review at the *Journal of Operations Management* [2]–[4].

JOURNAL PUBLICATIONS

- [1] **Zhou, C.***, Ding, X., Chen, W., Lei, L., Norrell, J., Tray, A., & Evens, A.M. (2026). “Enhancing efficiency and workflow in oncology outpatient services through simulation-based optimization.” *Annals of Operations Research*, 362(1), 457–482. doi.org/10.1007/s10479-026-07181-2 (Dissertation, opening chapter)

An on-site laboratory combined with optimized scheduling reduces average visit length by 75–90 minutes (roughly 30 percent) at an NCI-designated cancer center; joint implementation dominates any single change.

MANUSCRIPTS UNDER REVIEW AND IN REVISION

Author order reflects true submission order; * marks the corresponding author.

- [2] **Zhou, C.***, Lin, T., Ding, X., & Chen, W. “Balancing Workforce Fissuring and Service Quality: Evidence from Dialysis Operations.” **Under Review (3rd round)**, *Journal of Operations Management* (Special Issue on the Fissured Workforce Economy). (Dissertation ch. 1; job market paper)
- [3] **Zhou, C.**, Dreyfus, D.*, & Bagchi, A. “Exploring the Triple Aim Through Clinician Empowerment: The Role of Resources and Health Disparities on Quality Outcomes.” **Under Review (2nd round)**, *Journal of Operations Management*. (Dissertation ch. 2)
- [4] **Zhou, C.***, & Li, W. “The Misplaced Intelligence: AI Position and Output in Sequential Care Lines.” Under Review, *Journal of Operations Management*. (Dissertation ch. 3)
- [5] Dreyfus, D., **Zhou, C.***, & Lin, T. “Enhancing Clinical Quality of Care in Socially Disadvantaged Communities: Chain Affiliation and Dynamic Capabilities.” **Under Review (revision)**, *Health Care Management Science* (Special Issue on Underserved Populations).
- [6] **Zhou, C.***, & Li, W. “When Stars Fall: Quality Instability, Consumer Choice, and the Demand for Long-Term Care.” Under Review, *Decision Sciences*.
- [7] **Zhou, C.***, Li, W., & Alshalfan, A. “When the Fix Is Only Halfway: Sustainable Waste Routing and Toxic Releases.” **Under Revision**, *Production and Operations Management*.

Invited

- [8] **Zhou, C.***, & Li, W. “When the Choice Is Removed: Leased Owner-Operator Reliance, Mandated Reclassification, and Motor Carrier Safety Performance.” **Full manuscript invited, second stage**, *Journal of Business Logistics*, Special Topic Forum on AI-Integrated Research.

Recently Submitted

- [9] **Zhou, C.*** “When Customer Turnover Becomes Constrained: Trade-Policy Turbulence and Firm Innovation.” *Production and Operations Management*.
- [10] **Zhou, C.***, & Li, W. “When Today’s Performance Becomes Tomorrow’s Benchmark: Dynamic Incentives in Healthcare Value-Based Purchasing.” *Management Science*.

SELECTED WORKING PAPERS

- [11] **Zhou, C.*** “Beyond the Turning Point: The Height, Width, and Location of Curvilinear Effects in OM.” Manuscript complete, 42 pp.; census of 115 published studies.
- [12] **Zhou, C.*** “Deep Reinforcement Learning for Forward-Looking Chemotherapy Appointment Scheduling.” Manuscript complete.
- [13] **Zhou, C.**, Li, W.*, & Rogers, D. “Extension Services, Market Remoteness, and Agricultural Productivity:

HONORS, FELLOWSHIPS, AND RESEARCH FUNDING

Summer Research Grant (\$4,000 annually, four years)	\$16,000	2022–2025
Alfred J. Battaglia Memorial Fellowship (\$3,500, awarded twice)	\$7,000	2024–2025
Graduate School Academic Excellence Award	\$1,000	2025

Personal fellowships and awards to date: \$24,000.

Funded Project Participation

“Workflow Optimization and Structure Benchmark,” Rutgers Cancer Institute of New Jersey (X. Ding, PI; graduate researcher on the modeling and simulation team) \$105,767 2022–2023

Supported the oncology outpatient workflow study published in *Annals of Operations Research* [1].

DATABASES AND RESEARCH INFRASTRUCTURE

Panels assembled and validated from public and administrative sources; each supports several papers and is ready for new projects from the first month of an appointment.

- **Diagnostic-AI adoption panel** — hospital-year adoption and process position of diagnostic AI in coronary care pathways, built from Medicare provider-level billing records (NPI $\times$ HCPCS $\times$ year) linked to HCRIS cost reports; underlies [4]
- **Buyer-supplier relationship and innovation database** — supply chain ties built from statutory SEC 10-K/20-F disclosures (buyer-disclosed suppliers united with suppliers’ major-customer disclosures), matched to CIK/GVKEY by automated lookup with manual audit and linked to KPSS patent records through CRSP-Compustat; 3,418 buyers, 21,651 buyer-years, FY2012–2023; underlies [9]
- **U.S. dialysis facility panel** — facility-year staffing, contract-labor, and CMS quality records covering the full sector; underlies [2] and [5]
- **CMS long-term care panel** — ownership transitions, staffing records, and star-rating histories; underlies [6]
- **Motor carrier safety and classification records** — carrier-level safety and driver-classification histories; underlies [8]
- **U.S. airline route-level operations** — forty years of route-level operating records
- **Rutgers Cancer Institute of New Jersey** (2022–2024, NCI-designated) — patient-flow and infusion-center scheduling data; simulation-optimization models adopted for deployment; underlies [1] and [12]

TEACHING EXPERIENCE

Instructor of Record (sole instructor)

2025–2026	29:799:310	Demand Planning & Fulfillment	Newark
	52:620:326	Supply Chain Analytics	Camden
2026–2027	29:799:310	Demand Planning & Fulfillment	Newark
	33:799:310	Demand Planning & Fulfillment	New Brunswick

Course Evaluations (Demand Planning & Fulfillment, Rutgers–Newark)

Most recent (Summer 2026): overall teaching effectiveness 4.78/5.00 (school 4.21); course quality 4.67/5.00 (school 4.14); 9 of 17 enrolled students responding.

Prior term (Summer 2025): overall teaching effectiveness 4.91/5.00 (school 4.33; 98th percentile); course quality 5.00/5.00; response rate 85%.

Full SIRS reports available as supplemental material.

Course Development and Pedagogical Innovation | www.chenhaozhou.me/teaching.html 

- **Business Analytics & Machine Learning course platform** — modular curriculum, 10+ modules, three capstone projects, Python labs
- **17 browser-based operations simulations** — M/M/c queue, newsvendor, EOQ, vehicle routing, facility location, port management with AI recommendations, reverse factoring, VaR
- **Four coding reference guides** (Python, Stata, R, SQL) and **six student handbooks**, including an open-access guide to endogeneity and causal inference

Teaching Assistant (selected)

Undergraduate

- Demand Planning & Fulfillment (2021–2025)
- Process & Operations Management (2024)
- Supply Chain Analytics Essentials (2023)
- Six Sigma and Lean Manufacturing (2023)

Graduate

- Supply Chain Finance (2024)
- Global Procurement & Supply (2023–2024)
- Supply Chain Analytics (2023)
- Operations Analysis (2023)
- Project Management (2022)

Certification Instruction and Examination Support

Lean Six Sigma Green Belt Technical Certificate, Rutgers Business School

2022–2025

Advanced Certification in Procurement, Rutgers Business School

2025

CONFERENCES AND ACADEMIC SERVICE

Conference Proceedings

- **Zhou, C.** (2021). “House price prediction using polynomial regression with Particle Swarm Optimization.” *Journal of Physics: Conference Series*, 1802(3), 032034. doi.org/10.1088/1742-6596/1802/3/032034 
- **Zhou, C.** (2020). “Quantitative investment strategy analysis based on machine learning for share dealing.” *2020 7th International Conference on Information Science and Control Engineering (ICISCE)*, 1051–1057. doi.org/10.1109/ICISCE50968.2020.00215 

Conference Presentations

- **MSOM Annual Conference, Harvard Business School (2026)** — two papers invited, after review, into the final 36 presentations; both delivered July 2026
- **DSI Annual Meeting (2026)** — two papers accepted to the full-paper track
- POMS Annual: “Community Health Disparities and Dialysis Performance” (2026)
- POMS Annual: “Team Continuity Amid Workforce Fissure” (2026)
- DSI Doctoral Research Showcase: “Healthcare Operating in a Fissured Workforce” (2025)
- INFORMS Service Science: “Resilience in Health Care Operations” (2025)

Session Leadership

- Session Chair, POMS Healthcare Analytics (2025, 2026); INFORMS Service Science (2025)

Review Service

- *Annals of Operations Research* (2025–present); *Health Care Management Science* (2024–present); *Technology and Health Care*
- Decision Sciences Institute annual conference — Project Management, Gig Economy (2025); Healthcare (2024)

Professional Memberships

AOM | DSI | IEEE | INFORMS | POMS

BOOKS

Hui, J., Liu, J., **Zhou, C.**, & Zhou, W. (Eds.) (2025). *Strategies for Enhancing the Effectiveness of Internal Audit and Risk Control* (Chapters I, II, IX–XI). China Modern Economic Publishing House. ISBN 978-7-5119-3474-1. Adopted as a course textbook, Shanxi University of Finance and Economics.

REFERENCES

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